

MTH 204 : Lecture 16

Series Solution:

- The power series method can be applied to linear ODEs with variable coefficients such as

$$y'' + p(x)y' + q(x)y = 0$$

because the coefficients p and q can also be substituted by a power series

Ex: $(1-x^2)y'' - 2xy' + 2y = 0$

Let $y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots$

Then $y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + \dots$

and $y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \dots$

So, $(1-x^2)y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \dots$

$- 2a_2x^2 - 6a_3x^3 - 12a_4x^4 + \dots$

$- 2xy' = -2a_1x - 4a_2x^2 - 6a_3x^3 - 8a_4x^4 - \dots$

$2y = 2a_0 + 2a_1x + 2a_2x^2 + 2a_3x^3 + 2a_4x^4 + \dots$

So, $2a_2 + 2a_0 = 0 \Rightarrow a_2 = \boxed{-a_0}$

$6a_3 = 0 \Rightarrow a_3 = \boxed{0}$

$12a_4 - 4a_2 = 0 \Rightarrow a_4 = \frac{1}{3}a_2 = \boxed{-\frac{1}{3}a_0}$

$\vdots$

So, the general solution can be written as:

$y = a_1x + a_0 - a_0x^2 - \frac{1}{3}a_0x^4 - \dots$

$\Rightarrow y = a_1x + a_0 \left(1 - \frac{x^2}{1} - \frac{1}{3}x^4 - \dots \right) = \boxed{a_1x + a_0 \left(1 - \sum_{m=1}^{\infty} \frac{x^{2m}}{(2m-1)} \right)}$

Ex: $y'' - y' + xy = 0$

Let $y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots$

Then $y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + \dots$

$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \dots$

Then $-y' = -a_1 - 2a_2x - 3a_3x^2 - 4a_4x^3 - 5a_5x^4 - \dots$

$xy = a_0x + a_1x^2 + a_2x^3 + a_3x^4 + \dots$

So, $2a_2 - a_1 = 0 \Rightarrow a_2 = \frac{1}{2}a_1$

$6a_3 - 2a_2 + a_0 = 0 \Rightarrow 6a_3 = 2a_2 - a_0 = a_1 - a_0$

$\Rightarrow a_3 = \left[\frac{1}{6}a_1 - \frac{1}{6}a_0 \right]$

$12a_4 - 3a_3 + a_1 = 0 \Rightarrow 12a_4 = 3a_3 - a_1 = \frac{1}{2}a_1 - \frac{1}{2}a_0 - a_1$

$\Rightarrow a_4 = -\frac{1}{24}(a_1 + a_0)$

$20a_5 - 4a_4 + a_2 = 0 \Rightarrow 20a_5 = 4a_4 - a_2$

$\Rightarrow 20a_5 = 4a_4 - a_2 = -\frac{1}{6}a_1 - \frac{1}{6}a_0 - \frac{1}{2}a_1 = -\frac{(4a_1 + a_0)}{6}$

$\Rightarrow a_5 = -\frac{1}{120}(4a_1 + a_0)$

$30a_6 - 5a_5 + a_3 = 0 \Rightarrow 30a_6 = 5a_5 - a_3$

$\Rightarrow 30a_6 = -\frac{1}{6}a_1 - \frac{1}{24}a_0 - \frac{1}{6}a_1 + \frac{1}{6}a_0$

$\Rightarrow 30a_6 = -\frac{1}{3}a_1 + \frac{1}{8}a_0 = \frac{3a_0 - 8a_1}{24} \Rightarrow a_6 = \frac{3a_0 - 8a_1}{720}$

So, $y = \left(a_0 - \frac{1}{6}a_0x^3 - \frac{1}{24}a_0x^4 - \dots \right) + \left(a_1x + \frac{1}{2}a_1x^2 + \frac{1}{6}a_1x^3 - \frac{1}{24}a_1x^4 - \dots \right)$

$= a_0 \left(1 - \frac{1}{6}x^3 - \frac{1}{24}x^4 - \dots \right) + a_1 \left(x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{24} - \dots \right)$

Operations on Power Series:

(1) Termwise addition: Two power series may be added term by term. More precisely: if the series $\sum_{m=0}^{\infty} a_m(x-x_0)^m$ and $\sum_{m=0}^{\infty} b_m(x-x_0)^m$ have positive radius of convergence and their sums are $f(x)$ and $g(x)$, then the series $\sum_{m=0}^{\infty} (a_m+b_m)(x-x_0)^m$ converges and represents $f(x)+g(x)$ for each x that lies in the interior of the convergence interval common to each of the two series

(2) Termwise Multiplication: Two power series may be multiplied term by term. More precisely: Suppose the two series in (1) have positive radius of convergence and let $f(x)$ and $g(x)$ be their sums. Then the series obtained by multiplying each term of the first series by each term of the second series and collecting like powers of $(x-x_0)^m$, that is,

$$a_0b_0 + (a_0b_1 + a_1b_0)(x-x_0) + (a_0b_2 + a_1b_1 + a_2b_0)(x-x_0)^2 + \dots$$
$$= \sum_{m=0}^{\infty} (a_0b_m + a_1b_{m-1} + \dots + a_mb_0)(x-x_0)^m$$

converges and represents $f(x)g(x)$ for each x in the interior of the convergence interval of each of the two given series.

(4) Vanishing of all coefficients (Identity Theorem for Power Series)

If a power series has a positive radius of convergence and a sum that is identically zero throughout its interval

of convergence, then each coefficient of the series must be zero.

(4) Termwise Differentiation: A power series may be differentiated term by term. More precisely:

differentiated term by term. More precisely:

If $y(x) = \sum_{m=0}^{\infty} a_m (x-x_0)^m$ converges for $|x-x_0| < R$ where $R > 0$, then the series obtained by differentiating term by term also converges for those x and represents the derivative y' of y for those x :

$$y'(x) = \sum_{m=1}^{\infty} m a_m (x-x_0)^{m-1} \quad (|x-x_0| < R)$$

Similarly for the second and further derivatives.

Legendre's Equation:

$$(1-x^2)y'' - 2xy' + x(x+1)y = 0 \quad \dots \textcircled{1}$$

We can transform this equation into

$$y'' - \frac{2x}{1-x^2} y' + \frac{n(n+1)}{1-x^2} y = 0$$

The coefficient functions are differentiable at $x=0$ but not at $x=\pm 1$

Thus we can use a power series

around 0 as a solution.

$$\text{Let } y = \sum_{m=0}^{\infty} a_m x^m, \quad y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$$

$$y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$$

$$\text{Let } k = n(n+1)$$

Substituting in (1), we get

$$(1-x^2) \left(\sum_{m=2}^{\infty} m(m-1) a_m x^{m-2} \right) - 2x \left(\sum_{m=1}^{\infty} m a_m x^{m-1} \right) + k \left(\sum_{m=0}^{\infty} a_m x^m \right) = 0$$

$$\Rightarrow \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2} - \sum_{m=2}^{\infty} m(m-1) a_m x^m - \sum_{m=1}^{\infty} 2m a_m x^m + \sum_{m=0}^{\infty} k a_m x^m = 0$$

Let us substitute $m-2 = s$ in the first summation
 $m = s+2$

and $m=s$ in the other summations.

$$\sum_{s=0}^{\infty} (s+2)(s+1)a_{s+2}x^s - \sum_{s=2}^{\infty} s(s-1)a_s x^s - \sum_{s=1}^{\infty} 2sa_s x^s + \sum_{s=0}^{\infty} ka_s x^s = 0$$

Coefficient of x^0 :

$$2 \times 1 \times a_2 + ka_0 = 0 \Rightarrow 2a_2 + ka_0 = 0$$

Coefficient of x^1 :

$$3 \times 2 \times a_3 - 2a_1 + ka_1 = 0$$

$$\Rightarrow 6a_3 + (k-2)a_1 = 0$$

Coefficient of x^s :

$$(s+2)(s+1)a_{s+2} + (-s^2 + s - 2s + k)a_s = 0$$

$$\Rightarrow (s+2)(s+1)a_{s+2} - (s^2 + s - k)a_s = 0$$

$$\text{Now, } a_2 = -\frac{k}{2} a_0 = -\frac{n(n+1)}{2} a_0$$

$$a_3 = -\frac{(k-2)}{6} a_1 = -\frac{\{n(n+1)-2\}}{6} a_1$$

$$\Rightarrow a_3 = -\frac{(n^2+n-2)}{6} a_1$$

$$\Rightarrow a_3 = -\frac{(n+2)(n-1)}{6} a_1$$

In general,

$$a_{s+2} = \frac{s^2+s-k}{(s+2)(s+1)} a_s$$

$$= \frac{s^2+s-n(n+1)}{(s+2)(s+1)} a_s$$

$$= \frac{s^2 + (n+1)s - ns - n(n+1)}{(s+2)(s+1)} a_s$$

$$a_{s+2} = \frac{(s+n+1)(s-n)}{(s+2)(s+1)} a_s$$

$$\Rightarrow a_{s+2} = - \frac{(n-s)(n+s+1)}{(s+2)(s+1)} a_s$$

for $s=0, 1, 2, \dots$

(Recursion formula)

Let $s=2$:

$$a_4 = - \frac{(n-2)(n+3)}{4 \cdot 3} a_2$$

$$= \left\{ - \frac{(n-2)(n+3)}{4 \cdot 3} \right\} \left\{ - \frac{n(n+1)}{2!} \right\} a_0$$

$$= \frac{(n-2)n(n+1)(n+3)}{4!} a_0$$

Let $s=3$

$$a_5 = - \frac{(n-3)(n+4)}{5 \cdot 4} a_3$$

$$= \left\{ - \frac{(n-3)(n+4)}{5 \cdot 4} \right\} \left\{ - \frac{(n+2)(n-1)}{3!} a_1 \right\}$$

$$= \frac{(n-3)(n-1)(n+2)(n+4)}{5!} a_1$$

Thus inserting these expressions for the coefficients in $y = \sum_{n=0}^{\infty} a_n x^n$, we obtain

that there are actually two independent solutions and the general solution in $(-1,1)$ is a linear combination of the two:

$$y = a_0 y_1 + a_1 y_2$$

$$\text{where } y_1 = \left\{ 1 - \frac{n(n+1)}{2!} x^2 + \frac{(n-2)n(n+1)(n+3)}{4!} x^4 - \dots \right\}$$

$$\text{and } y_2 = \left\{ x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!} x^5 - \dots \right\}$$

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(Both the series converges in $(-1,1)$)

Since $\frac{y_1}{y_2}$ is not a constant,

y_1 and y_2 are linearly independent.

So, $y = a_0 y_1 + a_1 y_2$ is a general solution of the ODE on the interval $(-1, 1)$

Polynomial solution:

Consider the recursion relation:

$$a_{s+2} = - \frac{(n-s)(n+s+1)}{(s+2)(s+1)} a_s$$

If n is a positive integer then,

For $s=n$: $a_{s+2} = 0$

$$\Rightarrow 0 = a_{n+2} = a_{n+4} = a_{n+6} = \dots$$

If n is even, y_1 is a polynomial of degree n .

If n is odd, y_2 is a polynomial of degree n .

These polynomials will be referred to as $P_n(x)$ (Legendre Polynomial)

We will choose the highest coefficient (of degree n) to be

$$\left. \begin{aligned} a_n &= \frac{(2n)!}{2^n (n!)^2} && \text{if } n \text{ is a positive integer} \\ &= 1 && \text{if } n=0 \end{aligned} \right\}$$

From the recursion relation,

$$a_s = - \frac{(s+2)(s+1)}{(n-s)(n+s+1)} a_{s+2} \quad (\text{for } s \leq n-2)$$

we get

$$a_{n-2} = - \frac{n(n-1)}{2(2n-1)} a_n$$

$$= - \frac{n(n-1)}{2(2n-1)} \frac{(2n)!}{2^n (n!)^2}$$

$$\begin{aligned}
 &= - \frac{\cancel{n}(\cancel{n-1})}{2(2n-1)} \frac{(2n)!}{\cancel{n!} 2^n n!} \\
 &= - \frac{(2n)!}{2^n (n-2)! (n-1)! n \times 2 (2n-1)} \\
 &= - \frac{(2n-2)!}{2^n (n-2)! (n-1)!}
 \end{aligned}$$

Then similarly,

$$\begin{aligned}
 a_{n-4} &= - \frac{(n-2)(n-3)}{4(2n-3)} a_{n-2} = \frac{(n-2)(n-3)}{4(2n-3)} \frac{(2n-2)!}{2^n (n-2)! (n-1)!} \\
 &= \frac{(2n-4)!}{2^n 2! (n-2)! (n-4)!}
 \end{aligned}$$

In general, when
 $n - 2m \geq 0$,

$$a_{n-2m} = (-1)^m \frac{(2n-2m)!}{2^n m! (n-m)! (n-2m)!}$$

The resulting solution of Legendre's differential equation

is called the Legendre Polynomial of degree n and is denoted by $P_n(x)$. Hence,

$$P_n(x) = \sum_{m=0}^M (-1)^m \frac{(2n-2m)!}{2^n m! (n-m)! (n-2m)!} x^{n-2m}$$
$$= \frac{(2n)!}{2^n (n!)^2} x^n - \frac{(2n-2)!}{2^n 1! (n-1)! (n-2)!} x^{n-2} + \dots$$

where $M = \frac{n}{2}$ (if n is even)

$$= \frac{n-1}{2} \text{ (if } n \text{ is odd)}$$

Then,

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2} (3x^2 - 1)$$

$$P_3(x) = \frac{1}{2} (5x^3 - 3x)$$

$$P_4(x) = \frac{1}{8} (35x^4 - 30x^2 + 3)$$

$$P_5(x) = \frac{1}{8} (63x^5 - 70x^3 + 15x)$$

Note: The Legendre Polynomials $P_n(x)$

are orthogonal on the interval $[-1, 1]$

In fact,
$$\int_{-1}^1 P_i(x) P_j(x) dx = \frac{2}{2i+1} \delta_{ij}$$

So,
$$\int_{-1}^1 P_i(x) P_j(x) dx = 0 \quad \text{for } i \neq j$$

Note: $P_0(x) = 1$, $P_1(x) = x$

and we have

Bonnet's recursion formula

$$(n+1) P_{n+1}(x) = (2n+1)x P_n(x) - n P_{n-1}(x)$$

for $n=1, 2, \dots$

Definition: A function $f(x)$ is called analytic at a point $x=x_0$ if it can be represented by a power series in powers of $(x-x_0)$ with positive radius of convergence.

Regular and Singular points:

A regular point of the ODE

$$y'' + p(x)y' + q(x)y = 0$$

is a point x_0 at which $p(x)$ and $q(x)$ are analytic.

Similarly a regular point of the ODE

$$\tilde{h}(x)y'' + \tilde{p}(x)y' + \tilde{q}(x)y = 0$$

is a point x_0 at which $\tilde{h}, \tilde{p}, \tilde{q}$ are analytic and $\tilde{h}(x_0) \neq 0$ (so that we can divide by $\tilde{h}$)

- If x_0 is not a regular point, it is called a singular point.

Frobenius Method: Let $b(x)$ and $c(x)$ be any functions that are analytic at $x=0$ (i.e. they have no singularities at $x=0$)

Then the ODE $y'' + \frac{b(x)}{x} y' + \frac{c(x)}{x^2} y = 0$

has at least one solution that can be represented in the form

$$y = x^r \sum_{m=0}^{\infty} a_m x^m \quad (a_0 \neq 0)$$

where r may be any (real or complex) number (and r is chosen so that $a_0 \neq 0$)

The ODE also has a second solution (such that these two solutions are linearly independent) that may be similar to the previous one (with different r and coefficients) or may contain a logarithmic term.

Ex: Bessel's Equation:

$$y'' + \frac{1}{x} y' + \frac{x^2 - \nu^2}{x^2} y = 0$$

(Here $b(x) = 1$, $c(x) = x^2 - \nu^2$)

Ex: Hypergeometric Equation:

$$x(1-x)y'' + [c - (a+b+1)x]y' - aby = 0$$

(where a, b, c are constants)